Roblodex Platform Architecture and Delivery Proposal

A comprehensive proposal covering system design, phased delivery, technology choices, and investment structure for **Roblodex - a discovery and intelligence platform bringing semantic search, rich game understanding, and social signal layers to the Roblox ecosystem.**

Context

Roblodex is a discovery and intelligence platform combining:

Real-time Data Infrastructure

Continuous ingestion and live metrics powering the platform at its core.

Semantic Search and Classification

Intelligent understanding of content and user intent across the catalogue.

User-Driven Signal and Interaction

Behavioural signals that feed back into the system to improve over time.

The challenge is designing these systems to work together and scale from day one.

Our Understanding

Roblodex aims to become:

> The best way to discover and evaluate Roblox games

Winning depends on:

→ **Search that understands intent**

Going beyond keyword matching to surface what users are truly looking for.

→ **Rich and shareable game pages**

High-quality, information-dense pages that users want to return to and share.

→ **A system that improves with usage**

Every interaction makes the platform smarter and more valuable.

System Overview

Three core systems form the backbone of the Roblodex platform:

Data Engine

Continuous ingestion and metrics :: the live heartbeat of the platform, pulling and processing Roblox data at scale.

Meaning Machine

Taxonomy and semantic search:: the intelligence layer that understands what games are and what users want.

Signal Layer

Users, collections, interactions:: the feedback loop that makes the platform better the more it is used.

Key Principle

This product wins through:

- Search quality
- Experience clarity
- Speed to value

Not feature volume

The temptation in platform builds is to ship everything at once. Roblodex wins by doing fewer things exceptionally well; and building trust with users from the very first session.

Quality, clarity, and speed to value are the competitive advantages that compound over time.

Delivery Approach

We propose a staged approach that ensures early traction while maintaining long-term quality:

1. Discovery and Design Sprint

Define architecture and product experience before development begins.

2. Initial Release

A live, user-ready platform capable of onboarding users and generating traction.

3. Completion and Optimisation

Refinement, performance, and stability improvements to reach full quality.

This ensures early traction while maintaining long-term quality.

Discovery and Design Sprint

Duration

2 to 3 weeks

Investment

£12,500

Purpose: To define both architecture and product experience before development.

Includes:

- System architecture
- Roblox API validation
- Taxonomy and search design
- UX flows and UI direction
- Design of key screens
- Final roadmap

This is not a feasibility phase. The platform is fully buildable.

What Discovery Delivers

The Discovery and Design Sprint produces concrete, actionable outputs that enable faster and cleaner execution during build:

Approved designs for core product flows

Signed-off UX and UI direction ready for immediate development handoff.

Clear architecture and data model

A defined technical blueprint that removes ambiguity from the build phase.

Defined search approach

A validated strategy for semantic search and taxonomy that guides implementation.

Locked scope and milestones

A final roadmap with clear deliverables, timelines, and accountability.

This enables faster and cleaner execution during build.

Phase 1A: Initial Release

Duration

~12 weeks

Investment

£24,000

Outcome

A live, user-ready platform capable of onboarding users and generating traction.

Phase 1A is the core build — the moment Roblodex becomes real. By the end of this phase, the platform is live, functional, and ready for its first users.

What Ships in Initial Release

Phase 1A delivers a complete, functional platform. Every item below ships as part of the initial release:

- Core data ingestion system
- Functional semantic search
- High-quality game pages
- Discovery flows
- Roblox authentication
- Collections for saving and sharing
- SEO and AEO foundations

Phase 1B: Completion and Optimisation

Duration

~4 to 6 weeks

Investment

£12,000

Focus areas:

01

Search refinement and ranking improvements

Tuning the semantic search layer based on real usage data.

02

Performance optimisation

Ensuring the platform is fast and responsive at scale.

03

UX polish

Refining the user experience based on early feedback.

04

Stability and scaling improvements

Hardening the infrastructure for sustained growth.

Technology Architecture

The Roblodex stack is purpose-built for performance, intelligence, and scalability:

Frontend

Next.js

Server-side rendering and static generation for fast, SEO-friendly user interfaces.

API Layer

Node.js

High-performance API delivery powering product features and third-party integrations.

Data and Intelligence Layer

Python

Data ingestion, processing, and LLM-based classification for the Meaning Machine.

Technology Rationale

Each technology choice is deliberate. The combined approach enables both performance and intelligence:

Node.js

Powers fast product and API delivery - enabling rapid iteration and a responsive user experience from day one.

Python

Powers data ingestion and LLM-based classification - bringing intelligence and semantic understanding to the platform.

Combined approach enables both performance and intelligence.

Data and Infrastructure

PostgreSQL	Core relational data store
TimescaleDB	Time-series metrics and historical data
Typesense / Elasticsearch	Search indexing and retrieval
AWS or GCP	Cloud infrastructure and hosting
Queue-based pipelines	Reliable, scalable data ingestion

Roblox API Approach

Roblox APIs provide sufficient access for:

- Game metadata
- Player counts
- Discovery data

Handled through **batching**, **queue processing**, and a **tiered refresh strategy**.

This is an engineering consideration, not a feasibility risk.

Timeline

Weeks 1–3
Discovery and design

Weeks 4–15
Initial release

Weeks 16–20
Completion and optimisation

Cost Structure, Infrastructure, Risks, and Long-Term Opportunity

Total Investment: £48,500

£12.5K
Discovery and Design

£24k
Initial Release

£12K
Completion and Optimisation

Infrastructure Costs

Launch scale	£500 to £1,200 per month
Scaling	£1,500 to £4,000+ per month

Risks and Mitigation

Risk	Mitigation
API rate limits	Early validation in discovery
Search tuning complexity	Iterative refinement
LLM classification accuracy	Controlled scope

Long-Term Opportunity

Roblodex improves over time:

More users → more data

More data → better search

We can continue as a long-term partner to support growth, optimisation, and expansion.

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